In-Class Activity: PK-PD

Module 7b - PK-PD Relationship · In-class activity · 12 Oct 2026

Name _____ Date _____

1. A drug with an elimination half-life of 3 hours, apparent volume of distribution of 25 L, and minimum effective concentration of 1.2 mcg/mL is to be administered as a single 250 mg IV bolus dose.

1a What is the duration of action of this dose?

1b What dose would yield a duration of action of 12 hours?

2. The equation for concentration of drug in the plasma as a function of time following

a 500 mg IV bolus dose was found to be:

2a $C_p = 50e^{-0.2t}$ a. What is the duration of action of a single 500 mg IV bolus dose if the minimum effective concentration is 5 mcg/mL?

2b $C_p = 50e^{-0.2t}$ b. What dose would double the duration of action found in part a?

2c $C_p = 50e^{-0.2t}$ c. What is the expected change in the duration of action following a single 250 mg IV bolus dose instead of the 500 mg IV bolus dose?